



Algorithms Problem Solving

Level 3

Convert Line Data to Record



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Definition

Text Record Deserialization

-  Program converts one delimited text line back into an `sClient` structure.

-  This process reconstructs structured data from its serialized text representation.

Explanation

Token Splitting

-  The stored line is passed to `SplitString()` using the same delimiter used during serialization.

-  Each token represents one field from the original record.

Field Mapping

-  The vector positions are mapped to the corresponding structure members.

-  The order must match the order used by `ConvertRecordToLine()`.

Numeric Conversion

-  The balance field is converted from string to `double` using `stod()`.

-  Other fields that are already strings can be assigned directly.

Procedures

1. Receive the stored text line.
2. Split the line using ` "#//#" `.
3. Assign `vector[0]` to `AccountNumber`.
4. Assign `vector[1]` to `PinCode`.
5. Assign `vector[2]` to `Name`.
6. Assign `vector[3]` to `Phone`.
7. Convert `vector[4]` using `stod()` and assign it to `AccountBalance`.
8. Return the reconstructed record.

💡 Example

Line:

A150#/#1234#/#Ahmad#/#079#/#5000

➡ Result

Account Number: A150

Pin Code: 1234

Name: Ahmad

Phone: 079

Balance: 5000

Note

- ◆ `SplitString()` and `ConvertRecordToLine()` must use the same delimiter.
- ◆ The field order in the file is part of the record format.
- ◆ `stod()` converts a valid numeric string into a floating-point value.

⊘ Warning

- ◆ Accessing a vector index that does not exist causes an out-of-range error.
- ◆ ``stod()`` can throw an exception when the balance token is not a valid numeric value.
- ◆ Missing or malformed fields should be validated before assigning them to the structure.



Comparison

▨ `ConvertLineToRecord()`

- ✓ Performs deserialization.
- ✓ Converts text tokens into structured fields.
- ✓ Restores numeric types where necessary.

▤ `PrintClientRecord()`

- ✓ Displays an already reconstructed client.
- ✓ Does not parse or deserialize data.
- ✓ Focuses on presentation rather than conversion.

✦ Summary

- ✓ Splits a serialized client line into field tokens.
- ✓ Maps tokens back to struct members.
- ✓ Converts the balance from text to `double`.
- ✓ Reconstructs the original structured record.

Thank you 😊